FIT5145 FOUNDATIONS OF DATA SCIENCE

Demographic Ageing and Healthcare Infrastructure in China

A Temporal Analysis of Supply Response and Regional Mismatch

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Aging pressure is rising across China's 31 provinces — but **are hospitals keeping up**?

This study quantifies the infrastructure response lag using a **Distributed Lag Model** and constructs a provincial **Mismatch Index** to identify where capacity debt is accumulating.

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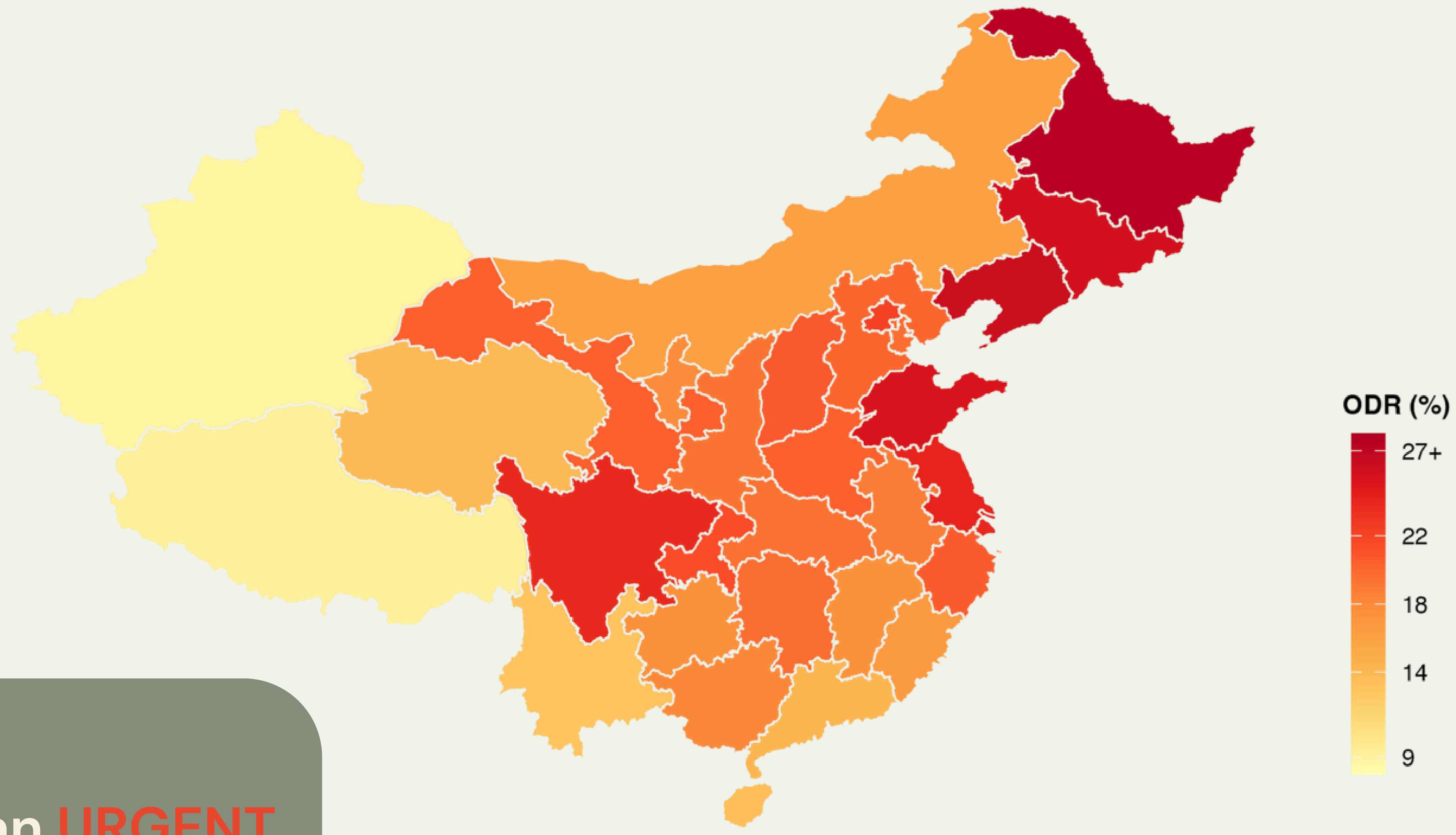
China is experiencing one of the world's fastest-growing aging populations

- **Rapid Aging:** In **2025**, about **23%** of China's population was over age 60, projected to reach **28%** by **2040** and potentially **39%** by **2050.**
- **Elderly Population Growth:** The number of people aged 60+ is expected to exceed **300 million** in **2024**, reaching **500 million** by **2048.**
- **80+ Demographic:** The 80-year-old and over age group is expanding, presenting unique challenges for **long-term care**

It is an URGENT issue faced by individuals and government

Elderly Dependency Ratio Across Chinese Provinces (2024)

Northeast & Sichuan Basin show the most severe aging pressure



15th Five-Year Plan (2026–2030) **designates 'Silver Economy' as a national priority**

Source: National Bureau of Statistics of China, 2024 Sample Survey
Note: ODR = Population aged 65+ / Population aged 15–64 × 100%

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Research Gap

Does my province have enough healthcare capacity to meet rising elderly demand?

Identify

Map which provinces face the most severe aging pressure using ODR data (2016–2024)

Demand & Supply

Quantify how aging pressure affects outpatient demand and hospital bed supply — and how long the adjustment takes

Predict and reallocate

Construct a Mismatch Index to identify provinces carrying a capacity debt entering the 15th Five-Year Plan

Methods:
Panel regression · Distributed Lag Model (DLM) · Mismatch Index

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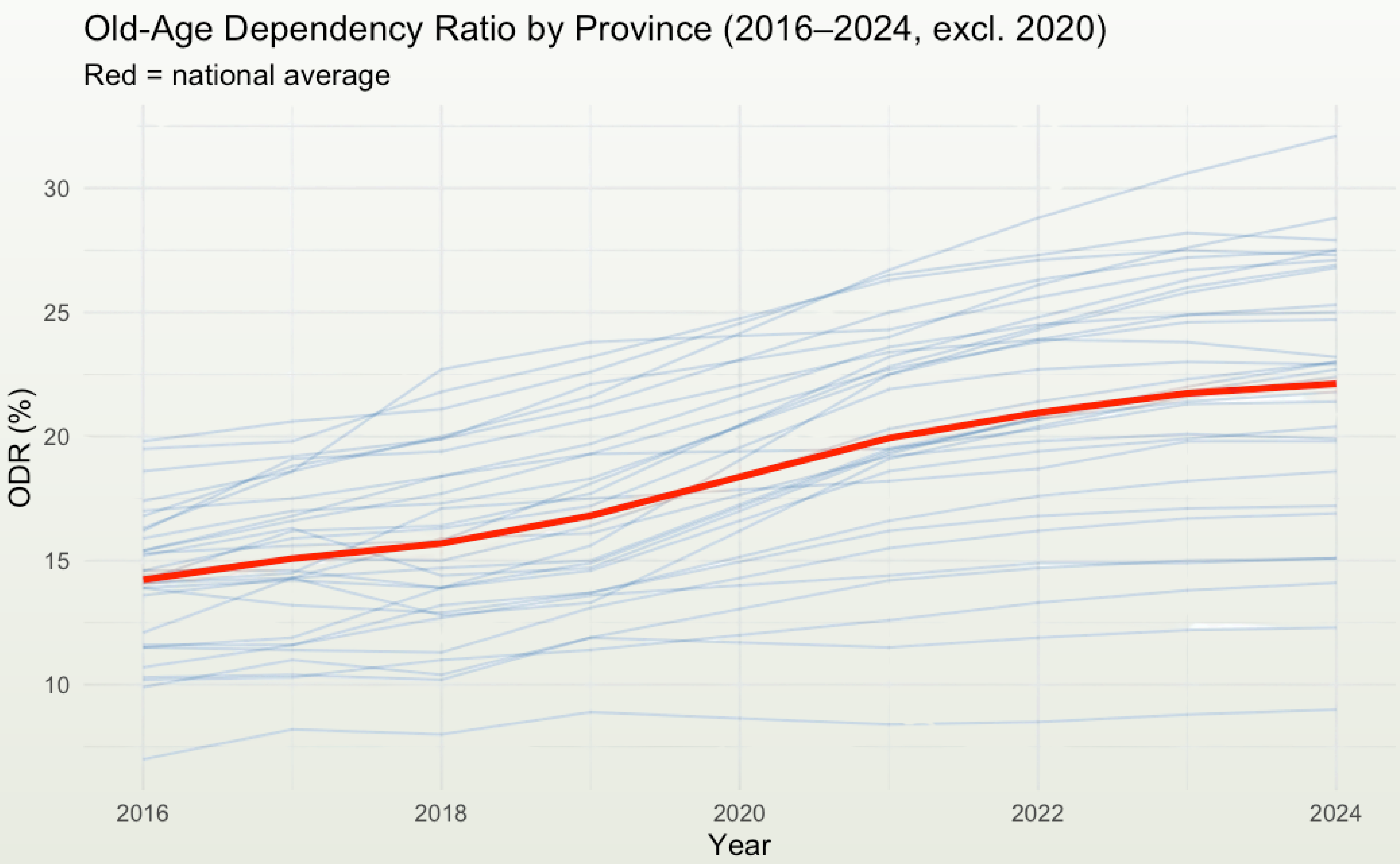
Data Overview



All provinces show rising ODR

the national average increased from ~14% in 2016 to ~21% in 2024, a 50% increase in just 8 years

The spread is widening

the highest-ODR provinces are aging nearly twice as fast as the lowest

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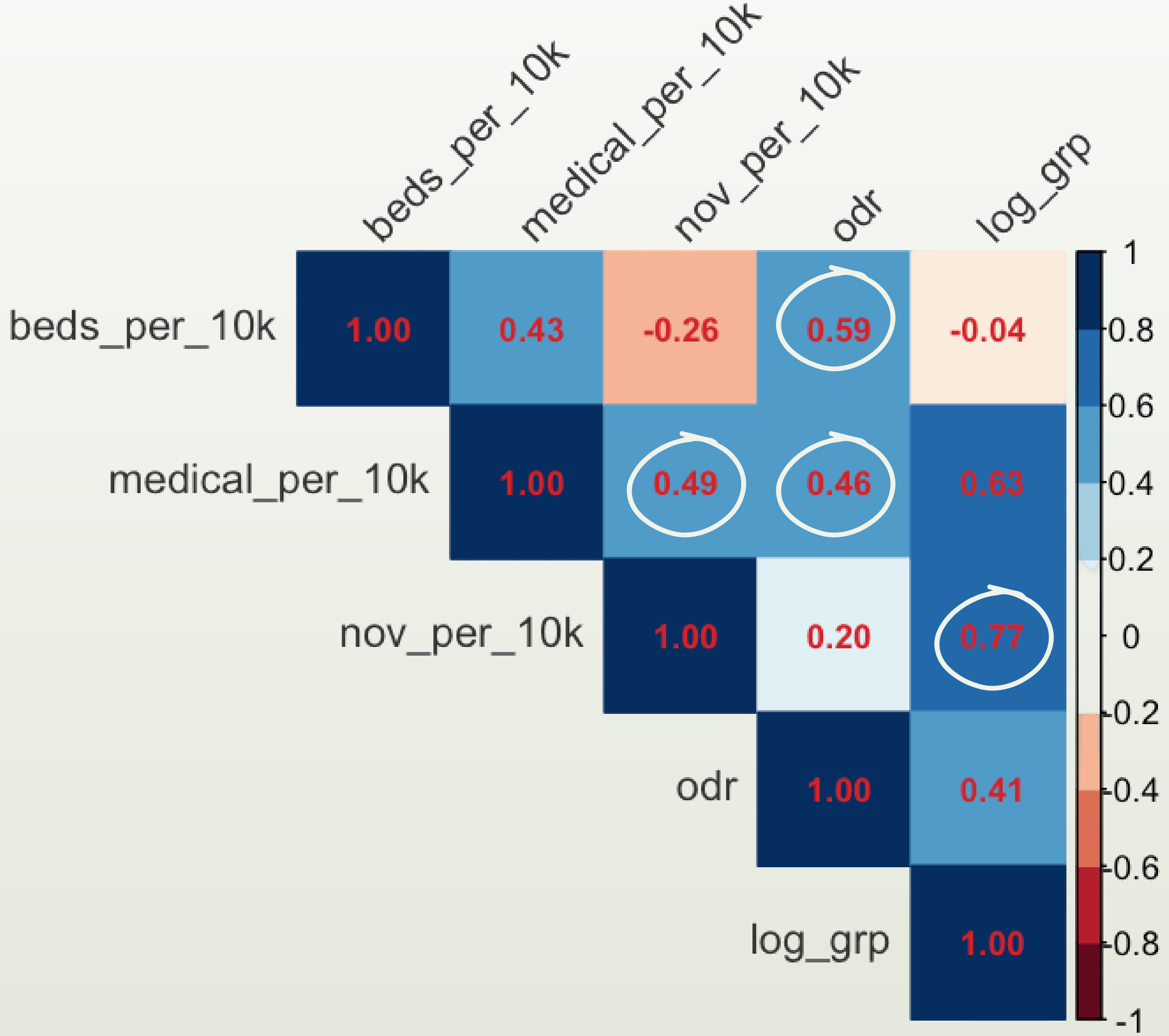
Data & Preliminary Evidence

Correlation Matrix: Healthcare & Demographic Variables



Key Metrics	Correlation Coefficient	Insights
Strong Economic Predictor	+ 0.77	Wealthier provinces generate significantly more outpatient activity
Demographic Pressure	+ 0.59 + 0.46	Higher ODR correlates with greater healthcare resource density — but the relationship is not immediate
Infrastructure Synergy	+ 0.43	Bed and personnel growth are moderately coupled but not perfectly aligned

The correlation calculation used use = "complete.obs", ensuring that the missing values identified in the nov variable (e.g., Tibet) do not distort the relationship coefficients.

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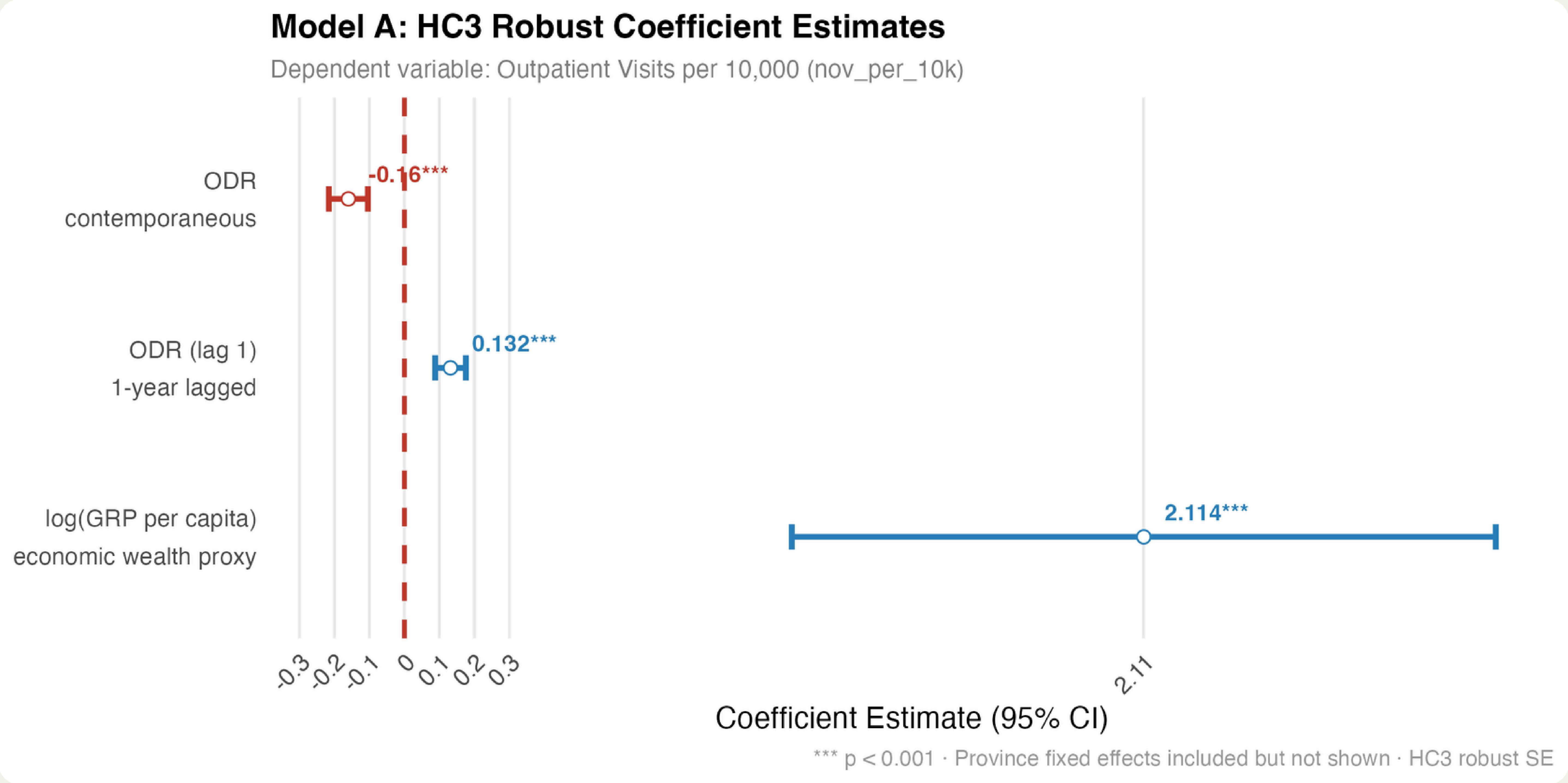
Demand responds with delay



Aging pressure does not immediately raise outpatient demand — the surge arrives one year later. Economic wealth remains the dominant driver of healthcare utilisation.

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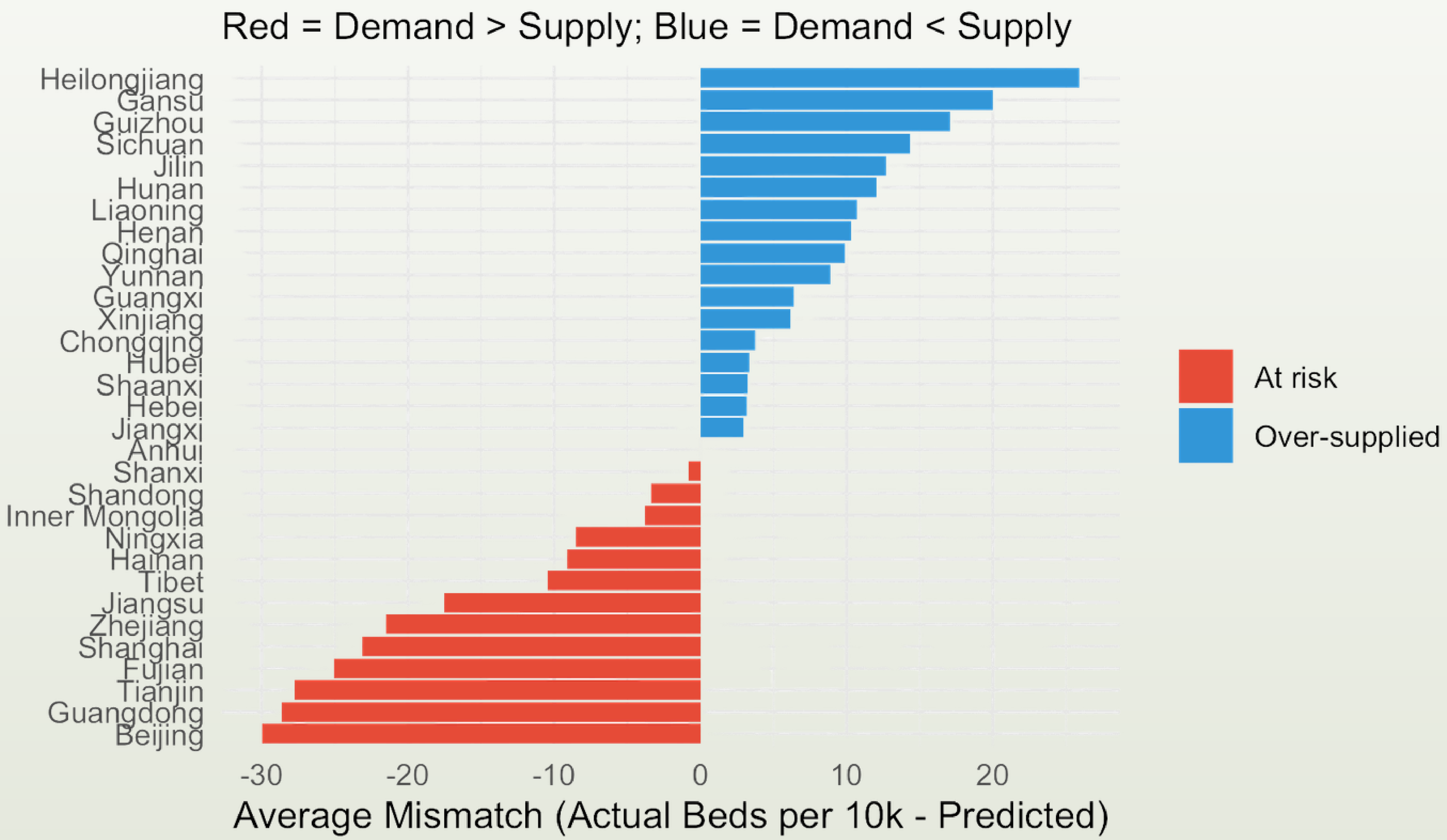
Supply mismatch exists



- Tier-1 economic hubs (Beijing, Guangdong, Shanghai) are most under-supplied — high patient inflow from other provinces inflates actual demand beyond local demographics
- Inland and northeastern provinces show relative surplus — likely reflecting population outflow reducing local demand
- Shanxi's gap is widening rapidly — a province to watch in the 15th FYP

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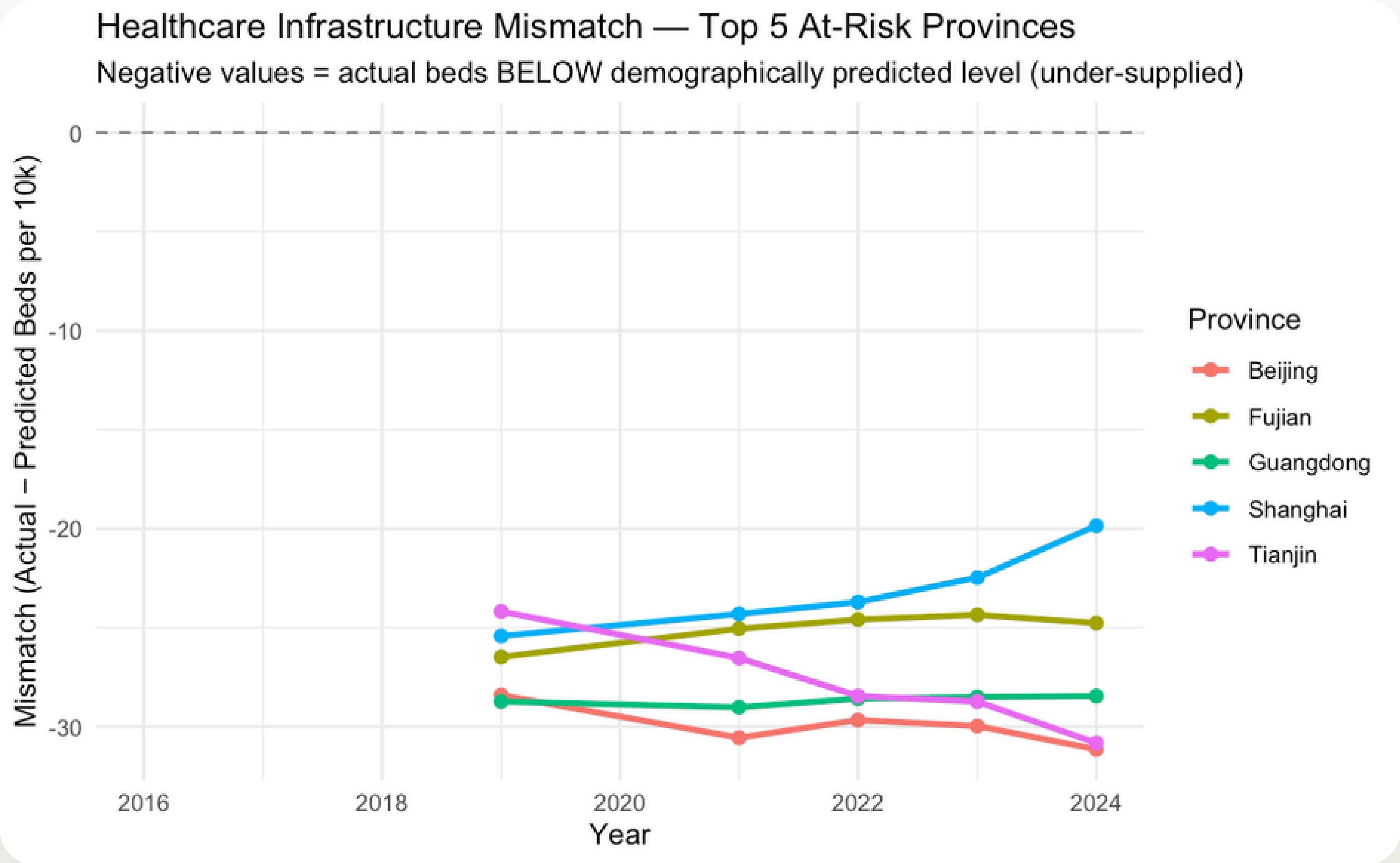
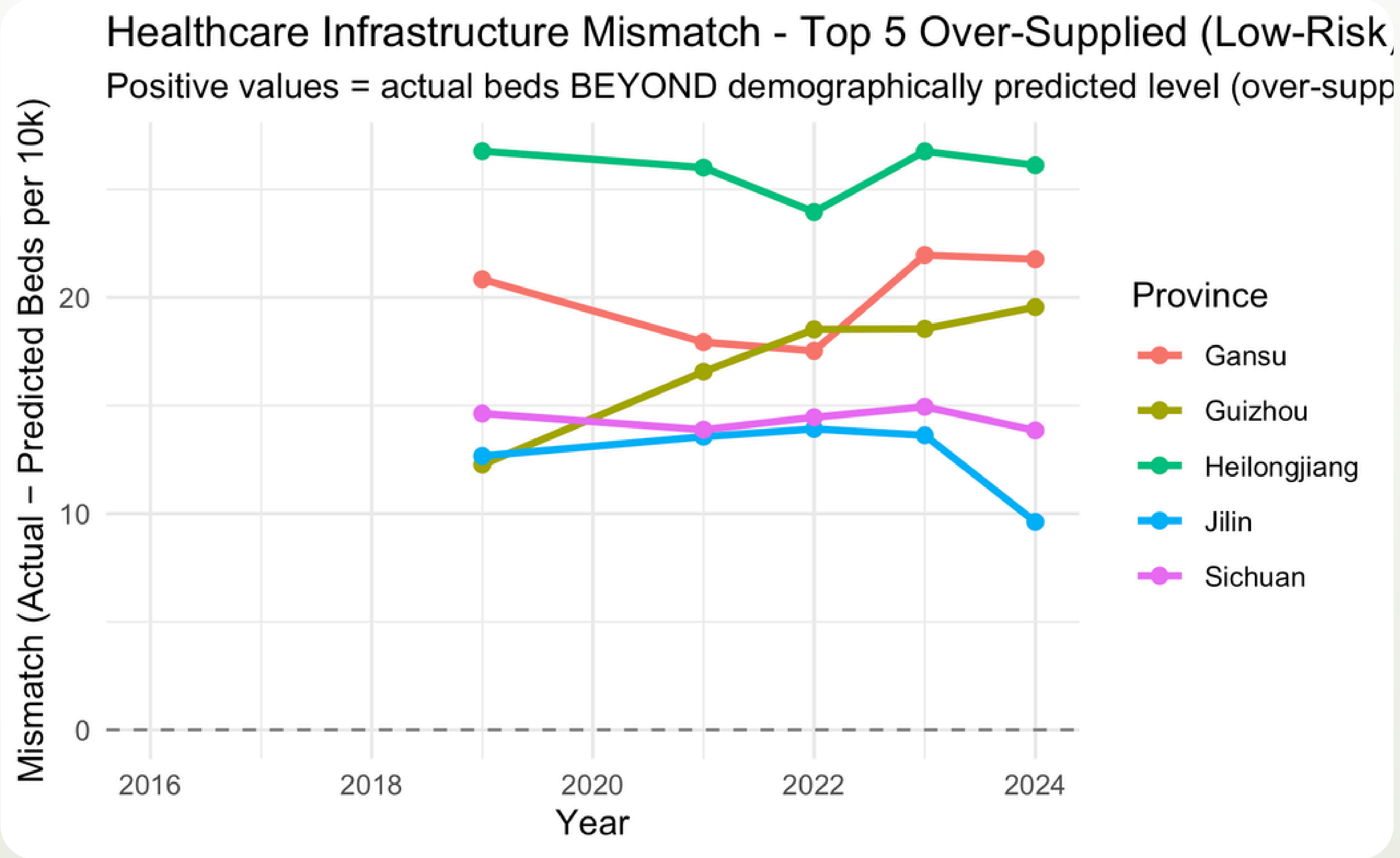
Who's at risk



Shanghai is actively closing its gap through aggressive investment. **Beijing** and **Tianjin** are diverging — inter-provincial patient migration amplifies local under-supply.

Who has surplus



Heilongjiang and **Gansu** exceed demographic benchmarks — but **Jilin's** declining surplus signals that aging pressure will eventually erode this buffer.

Policy Recommendations

🏥 **National Health Commission:** Prioritise 15th FYP infrastructure bonds for Beijing, Guangdong, Tianjin — provinces with the largest and widening capacity debt.

🏛️ **Provincial Planners**: Use the Mismatch Index as a fiscal priority map — lag-adjusted targets should account for demographic pressure 1–2 years ahead, not current snapshots.

💼 **Private Healthcare Investors:** Provinces where demand has decoupled from supply (Beijing, Shanghai, Fujian) represent the highest-yield PPP opportunity in the Silver Economy era.

Limitations & Conclusion

- Analysis is correlational, not causal — DLM identifies timing patterns but cannot establish causation
- Inter-provincial patient migration is not modelled — actual demand in hub cities is likely underestimated
- 2020 excluded — post-COVID recovery effects on healthcare utilisation not captured

China's healthcare infrastructure is not keeping pace with its demographic transition. Aging creates a one-year demand lag, and supply-side adjustment is even slower — leaving Tier-1 economic centres carrying a growing capacity debt.

As the 15th Five-Year Plan begins, this Mismatch Index provides a data-driven foundation for more responsive, demographically-aligned resource allocation.

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